and the population variance (1/N) $\sum_{\alpha} X_{\alpha}^{2} - p^2 = p - p^2 = pq$

The sample mean of the variable x, on the other hand, is $\frac{1}{n}\sum_{i=1}^{n} x_i = f/n$

Hence we find, on replacing $\bar{x}$ by f/n, μ by p and σ^2 by pq in the expressions E ($\bar{x}$) and σ_{x} given in preceding sections,

E (f/n) = p [in case of random sampling with replacement]

= p [in case of random sampling without replacement]

$\sigma_{f/n} = pq/\sqrt{n}$ [in case of random sampling with replacement]

$= \sqrt{\frac{pq}{n}\left(1 - \frac{N-n}{N-1}\right)}$ [in case of random sampling without replacement]

The comments made in connection with the standard error of the mean apply here also.

Check Your Progress 4

1) Discuss the meaning of random sampling with replacement and without replacement.

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2) Write the standard error of sample proportion.

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28.11 LET US SUM UP

In this unit, we have learnt the basic concepts of sampling theory like sample survey, its advantages and sample design. We also learnt about what kind of biases exist in a survey and various types of sampling methods e.g., purposive sampling, random sampling, stratified sampling and systematic sampling. Then we understood the concepts of parameter and statistic with a clear distinction between the two. We extended our understanding by finding out how the sampling distribution of a statistic needs to be worked out with dimensions like expectation and standard error of sample mean and sample proportion.

28.12 KEY WORDS

Cluster Sampling	:	The population is divided into clusters, ideally such that each cluster is as variable as the overall population (i.e., heterogeneity within clusters and homogeneity between clusters).
Quota Sampling	:	It attempts to obtain a representative sample by specifying quota controls on certain specified characteristics, such as age, gender, social class and any other variables relevant to the investigation being undertaken.
Population	:	In statistics, population is an aggregate of objects, animate or inanimate, under study. The population may be finite or infinite.
Purposive Sampling	:	Purposive sampling is one in which the sampling units are selected with definite purpose in view.
Random Sampling	:	A random sampling is one in which each unit of population has a known probability of being included in it.
Sample	:	A finite subset of statistical individuals in a population is called a sample and the number of individuals in a sample is called the sample size.
Sampling Distribution	:	A probability distribution consisting of all possible values of a sample statistic
Sampling Without Replacement	:	Once an element has been included in the sample, it is removed from the population and cannot be selected a second time
Sampling With Replacement	:	Once an element has been included in the sample, it is returned to the population. A previously selected element can be selected again and therefore may appear in the sample more than once.
Standard Error	:	The standard deviation of the sampling distribution of a statistic is known as its 'standard error', abbreviated as S.E
Stratified Sampling	:	Here the entire heterogeneous population is divided into a number of homogeneous groups, usually termed as *strata*, which differ from one another but each of these groups is homogeneous within itself.

Unbiased : A property of a point estimator that is present when the expected value of the point estimator is equal to the population parameter it estimates.

28.13 ANSWER OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) See Section 28.2
2) See Section 28.3
3) See Section 28.4

Check Your Progress 2

1) See Section 28.5
2) See Section 28.5
3) See Section 28.6

Check Your Progress 3

1) See Section 28.7
2) See Section 28.8
3) [Hint: n_1 =400 and n_2 =500, p_1 =300/400 =0.75, p_2 =300/500=0.6

   $p = (n_1p_1+n_2p_2)/(n_1+n_2) = 0.67$, $q = 1–p$; S. E $(p–p_1) = \sqrt{[((pq)/(n_1+n_2))(n_2/n_1)]} = 0.018$]

Check Your Progress 4

1) See Section 28.9
2) See Section 28.10

28.14 EXERCISES

Q1. A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Show that S.E. of the proportion of bad ones in a sample of this size is 0.022 and deduce that the percentage of bad pineapples in the consignment almost certainly lies between 8.5 and 17.5.

Ans. To estimate the proportion of spoiled pineapple in the whole consignment, we can use the sample proportion of defective pineapples in the sample.

The sample proportion of defective pineapples is given by:

$\hat{p} = x/n$ where x is the number of defective pineapples in the sample and n is the sample size.

$\hat{p} = 65/500 = 0.13$

To find the standard error of the estimate, we use the formula:

SE = $\sqrt{(\hat{p}(1-\hat{p})/n)}$

SE = $\sqrt{(0.13(1-0.13)/500)}$ = 0.022

We can use the sample proportion and standard error to construct a confidence interval for the true proportion of defective pineapples in the consignment. With a 95% confidence level, the interval is given by:

$\hat{p} \pm z^*SE$

where z is the critical value for the standard normal distribution at a 95% confidence level (z = 1.96)

Therefore, the 95% confidence interval for the true proportion of defective pineapples in the consignment is:

$\hat{p} \pm 1.96SE = 0.13 \pm 1.960.022 = [0.086, 0.174]$

This interval can be converted to a percentage interval by multiplying by 100. So, the interval becomes [8.6%, 17.4%]. We can therefore say with 95% confidence that the true proportion of defective pineapples in the consignment is between 8.6% and 17.4%.

It is almost certain that the percentage of bad pineapples in the consignment lies between 8.6% to 17.4%.

Q2. How does one get from a sample statistic an estimate of the population parameter?

Ans. A survey produces a sample statistic, which is used to estimate a population parameter. If you repeated the survey many times, using different samples each time, you might get a different sample statistic. And each of the different sample statistic would be an estimate for the *same* population parameter.If the statistic is unbiased, the average of all the statistics from all possible samples will be equal to the true population parameter; even though any individual statistic may differ from the population parameter

Q3. Random samples of size 225 are drawn from a population with mean 100 and standard deviation 20. Find the mean and standard deviation of the sample mean.

Ans. $\mu_{\bar{x}} = 100$, $\sigma_{\bar{x}} = 1.33$

Q4. Suppose the mean number of days to germination of a variety of seed is 22, with standard deviation 2.3 days. Find the probability that the mean germination time of a sample of 160 seeds will be within 0.5 day of the population mean.

Ans. 0.9940

Q5. Suppose speeds of vehicles on a particular stretch of roadway are normally distributed with mean 36.6 mph and standard deviation 1.7 mph.

a) Find the probability that the speed X of a randomly selected vehicle is between 35 and 40 mph.

b) Find the probability that the mean speed $\bar{x}$ of 20 randomly selected vehicles is between 35 and 40 mph.

Ans. a) 0.8036

b) 1.0000

Q6. Suppose that in a certain region of the country the mean duration of first marriages that end in divorce is 7.8 years, standard deviation 1.2 years. Find the probability that in a sample of 75 divorces, the mean age of the marriages is at most 88 years.

Ans. 0.9251

UNIT 29 SAMPLING DISTRIBUTIONS

Structure

29.0 OBJECTIVES

After going through this unit, you should be able to

- discuss the concept of sampling distribution of a statistic;
- enlist various properties of different types of sampling distribution;
- derive various results of different sampling distribution using Jacobian transformation;
- measure goodness of fit using Chi-square test; and
- analyse any sample when it is not randomly distributed.

29.1 INTRODUCTION

For a finite sample, it is not a big problem of assigning probabilities to the samples selected from the given population. However, where the sample size as well as the population is quite large, the number of all possible samples is also large. It becomes difficult to assign probabilities of a specified set of samples. Therefore, we must think of all possible ways of selecting the samples from the entire population. In this unit, we will learn about the sampling distributions with discrete and continuous population distributions, four fundamental distributions derived from the normal distribution and sampling distributions of mean and variance in random sampling from a normal distribution. In the end, we will also touch upon the central limit theorem.

29.2 CONCEPT OF SAMPLING DISTRIBUTION OF A STATISTIC

Sampling distribution of a statistic may be defined as the probability law, which the statistic follows, if repeated random samples of a fixed size are drawn from a specified population.

Let us consider a random sample $x_1, x_2, \ldots, x_n$ of size n drawn from a population containing N units. Let us further suppose that we are interested in the sampling distribution of the statistic $\bar{x}$ (i.e., sample mean), where $\bar{x} = \frac{1}{n}(x_1 + x_2 + \ldots + x_n)$.

If the population size N is finite, there is a finite number (say k) of possible ways of drawing n units in the sample out of a total of N units in the population. Although the k samples are distinct, the sample means may not be all different, but each of these will occur with equal probability. Thus, we can construct a table showing the set of possible values of the statistic $\bar{x}$ and also the probability that $\bar{x}$ will take each of these values. This probability distribution of the statistic $\bar{x}$ is called the 'sampling distribution' of sample mean. The above method is quite general, and the sampling distribution of any other statistic, say, median or standard deviation of the sample, may also be obtained.

If, however, the number (N) of units in the population is large, the number (k) of possible distinct samples being even larger, the above method of finding the sampling distribution cannot be applied. In this case, the values of $\bar{x}$ obtained from a large number of samples may be arranged in the form of relative frequency distribution. The limiting form of this relative frequency distribution, when the number of samples considered becomes infinitely large, is then called the 'sampling distribution of the statistic'. When the population is specified by a theoretical distribution (e.g., binomial or normal), the sampling distribution can be theoretically obtained. The knowledge of sampling distribution is necessary in finding 'confidence limits' for parameters and in 'testing statistical hypothesis'.

In this unit, we will highlight the various properties of different sampling distributions. We will mainly concentrate on how different sampling distributions work and in doing so we use several statistical formulae. Our aim is to present a theoretical overview of the topic. Here, what is important is to have the idea of topic theoretically and the numerical part will be covered subsequently.

29.3 THREE FUNDAMENTAL DISTRIBUTIONS DERIVED FROM NORMAL DISTRIBUTION

29.3.1 Chi-square Distribution

The square of a standard normal variate is known as a chi-square variate with 1 degree of freedom (df).

Thus, if $x \sim N(\mu, \sigma^2)$, then $z = \dfrac{x-\mu}{\sigma} \sim N(0,1)$

and $z^2 = \left(\dfrac{x-\mu}{\sigma}\right)^2$, is a chi-square variate with 1 df.

In general, if x_i, (i = 1, 2, ..., n) are n independent normal variates with mean μ_i and variance σ_i^2, (i = 1, 2, ..., n), then,

$$\chi^2 = \sum_{i=1}^{n}\left(\frac{x_i-\mu_i}{\sigma_i}\right)^2, \text{ is a chi-square variate with n df.}$$

It has the probability density function (p.d.f)

$$f(\chi^2) = \frac{1}{2^{n/2}\Gamma(n/2)} \exp\left[-x^2/2\right](\chi^2)^{\frac{n}{2}-1}$$

where $0 < \chi^2 < \infty$.

The p.d.f of x, the positive square root of χ^2, is immediately found to be

$$f(\chi) = \frac{1}{2^{(n-2)/2}\Gamma(n/2)} \exp\left[-\chi^2/2\right]\chi^{n-1}$$

where $0 < \chi < \alpha$.

For $n \le 2$ the density $f(\chi^2)$ steadily decreases as χ^2 increases, while for n >2 there is a unique maximum at $\chi^2 = n-2$. The distribution is thus always positively skewed. The curve of the distribution of χ^2 with df = 7 is as shown in Fig. 29.1.

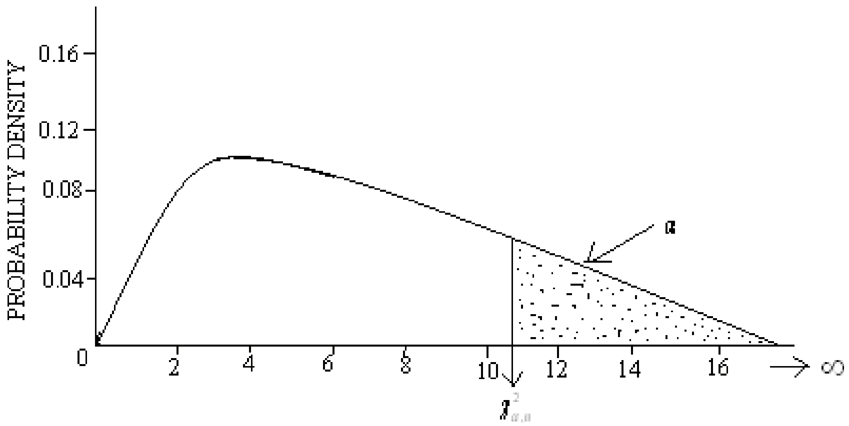

Fig. 29.1: χ^2 Distribution with 7 Degrees of Freedom (n = 7)

Consider, to begin with, the distribution of the square of a single standard normal variable, say, $z = x^2$.

z varies from 0 to ∞ and for 0 < a < b< ∞, noting that the transformation from x to z is two-to-one, we have,

$$p[a < z < b] = p\left[\sqrt{a} < x < \sqrt{b}\right] + p\left[-\sqrt{b} < x < -\sqrt{a}\right].$$

Thus, if g(z) is the p.d.f. of z and f(x) that of x, then

$$\int_a^b g(z)\,dz = \int_{\sqrt{a}}^{\sqrt{b}} f(x)\,dx + \int_{-\sqrt{b}}^{-\sqrt{a}} f(x)\,dx$$

$$= \int_a^b \left[f\left(\sqrt{z}\right)\frac{d}{dz}\left(\sqrt{z}\right) - f - \left(-\sqrt{z}\right)\frac{d}{dz}\left(-\sqrt{z}\right)\right] dz$$

(on putting $x = \sqrt{z}$ and $x = -\sqrt{z_i}$ in the first and the second integrals respectively).

Hence, $g(z) = \left[f\left(\sqrt{z}\right) + f\left(-\sqrt{z}\right)\right]\left|\frac{dx}{dz}\right|$

$$= 2 \times \frac{1}{\sqrt{2\pi}} \exp\left[-z/2\right] \times \frac{1}{2\sqrt{z}}$$

$$= \frac{1}{2^{1/2}\Gamma\left(1/2\right)} z^{-1/2} \exp\left(-z/2\right)$$

Thus, the earlier result given by $f\left(\chi^2\right)$ is seen to be true for n = 1 also. If it is then assumed to be true for n = 1, the p.d.f. of $u = \sqrt{\sum_{i=1}^{t} x_i^2}$ is from the expression given by $f\left(\chi^2\right)$,

$$\frac{1}{2^{(t-2)/2}\,\Gamma\left(t/2\right)} \exp\left[- - u^2/2\right] u^{t-1}, 0 < u < \infty,$$

and that of $\vartheta = \sqrt{x_{t+1}^2}$ is again form the expression given by $f\left(\chi^2\right)$,

$$\frac{2^{1/2}}{\Gamma(1/2)} \exp\left(-\vartheta^2/2\right), 0 < \vartheta < \infty$$

The joint p.d.f of u and ϑ is, then

$$\frac{1}{2^{(t-3)/2} . \Gamma(1/2)\Gamma(t/2)} . \exp\left[-(u^2 + \vartheta^2)/2\right] u^{t-1}$$

Now, make the one-to-one polar transformation:

$$\left.\begin{aligned} u &= u'\cos\theta \\ \vartheta &= u'\sin\theta \end{aligned}\right\} (0 < u' < \infty, 0 < \theta < \pi/2)$$

Then, $u' = \sqrt{u^2 + \vartheta^2} \left(\sqrt{\sum_{i=1}^{t+1} x_i^2} \right)$.

Also, the Jacobian of the transformation is

$$\begin{vmatrix} \frac{\partial u}{\partial u'} & \frac{\partial u}{\partial \theta} \\ \frac{\partial \vartheta}{\partial u'} & \frac{\partial \vartheta}{\partial \theta} \end{vmatrix} = u'$$

Hence, the joint p.d.f of u' and θ is

$$\frac{1}{2^{(t-3)/2}\Gamma\left(\frac{1}{2}\right)\Gamma(t/2)\exp\left[-(u')^2/2\right](u')^t \cos^{t-1}\theta}$$

$$0 < u' < \alpha, 0 < \theta < \pi/2$$

Since $2\int_0^{\pi/2} \cos^{t-1}\theta d\theta = B\left(1/2, t/2\right)$, the p.d.f of u' is

$$\frac{1}{2^{(t-1)/2}\Gamma\left(\frac{t+1}{2}\right)} \exp\left[\frac{-(u')^2}{2}\right](u')^t,$$

Hence, the p.d.f of u'^2 comes out to be

$$\frac{1}{2^{(t+1)/2}\Gamma\left(\frac{t+1}{2}\right)} \exp\left[-u'^2/2\right]\left((u')^2\right)^{(t-1)/2}$$

The results given by $f(\chi^2)$ thus holds for n = t + 1 if it is assumed to hold for n = t. Since it has been already shown to be valid for n = 1, by mathematical induction it is found to hold for all positive integral values of n.

An important result regarding χ^2 distribution is to be noted.

Let y_1 and y_2 be two independent variables distributed as χ^2, with df equal to n_1 and n_2, respectively. Then the sum $y_1 + y_2$ may be shown to be distributed in the same form with df = $n_1 + n_2$. This may be regarded because of the definition of χ^2, since $y_1 + y_2$ is the sum of squares of $n_1 + n_2$, mutually independent standard normal variables. A direct proof may be given by considering the joint distribution of y_1 and y_2 and deriving from it the joint distribution of $y = y_1 + y_2$ and q, which are such that

$$\sqrt{y_1} = \sqrt{y}\cos\theta$$

$$\sqrt{y_2} = \sqrt{y}\sin\theta \qquad \left(0 < y < \infty, \;\; 0 < \theta < \pi/2\right)$$

This property is designated as the additive property of χ_S^2.